



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F)

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TITLE OF THE PROJECT		THE MINI COMPILER FOR ARITHMETIC EXPRESSION	
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Abstract

The Mini Compiler for Arithmetic Expressions is a simplified compiler designed to demonstrate the fundamental concepts and phases of compiler design. The project focuses on processing arithmetic expressions containing operands, operators such as addition, subtraction, multiplication, and division, and parenthesis. The compiler accepts an arithmetic expression as input and performs lexical analysis to divide the input into meaningful tokens such as numbers, operators, and parentheses. The tokens are then passed to the syntax analyzer, which verifies whether the expression follows the grammar and checks the correct arrangement of operators and operands. The semantic analysis phase performs basic validity checks, such as ensuring that the expression is logically meaningful and identifying errors such as invalid operations. After successful analysis, the compiler generates an intermediate representation, such as three-address code, for the given expression. This representation breaks a complex expression into simpler instructions that can be easily understood and processed by the next stage of compilation. The system also provides error detection and reporting for invalid expressions, helping users identify syntax or semantic errors. The project demonstrates how a high-level arithmetic expression is transformed step by step into an intermediate form. It provides a practical understanding of important compiler concepts including tokens, lexical analysis, grammar, parsing, syntax trees, semantic analysis, intermediate code generation, and error handling. The mini compiler can serve as an educational model for understanding the internal working of real-world compilers.

Signature of the Guide

HOD